Exercise 6

On local maxima and minima (Local extrema)

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **Determine the local maximum and local minimum value of the following functions :**

(1) $y = x^2 + 4x + 6$

(2) $f(x) = 4 - 6x - x^2$

(3) *(2nd Session 2010)* $y = \frac{1}{3}x^3 - 9x + 2$

(4) *(2nd Session 2006)* $f(x) = 2x^3 + 3x^2 - 12x + 5$

(5) $f(x) = x^3 + 3x^2 - 9x - 7$

(6) $f(x) = (x-2)^3 + 5$

(7) $f(x) = \frac{1}{4}x^4 - x^3$

(8) $f(x) = (x-1)^4 + 2$

(9) $f(x) = x\left(1 - \frac{4}{3}x^2\right)$

(10) *(1st Session 2002)* $f(x) = x^4 - 2x^2$

(11) $f(x) = 3x^5 - 5x^3$

(12) $f(x) = x(x-3)^2 + 2$

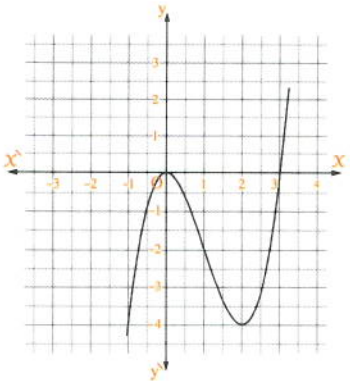
(13) $f(x) = \frac{1}{4}x^4 - 4x^3 - 2x^2 + 48x$

(14) $f(x) = 8x^4 - (x-2)^4$

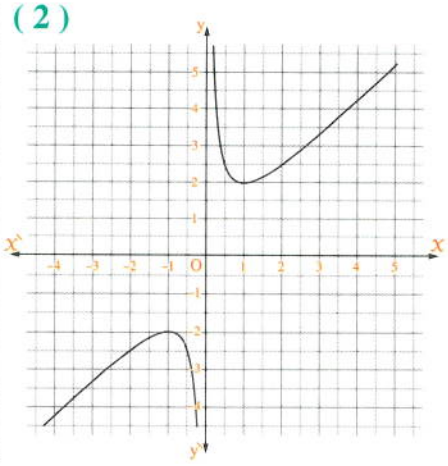
(15) $f(x) = x^4(x+2)^2$

2 **Determine the local maximum and local minimum value (if exist) of the following functions and state its type :**

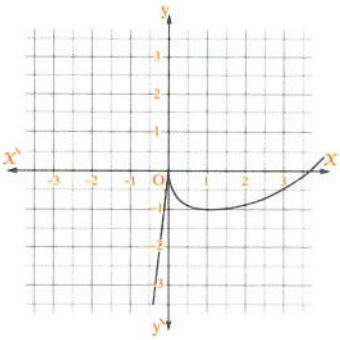
(1)



(2)



(3)



3 **Choose the correct answer from the given ones :**

(1) If the function f is a polynomial function and it has a local maximum value at the point (a, b), then $f'(a) =$ .....

(a) b (b) zero (c) $\frac{-b}{a}$ (d) undefined.

(2) The curve : $y = x^2 - 6x + 7$ has a local minimum value at $x =$ .....

(a) 2 (b) 3 (c) 4 (d) zero

(3) If the curve of the function f has $f(5) = 7$, $\dot{f}(5) =$ zero
$\ddot{f}(5) = -4$, then the point $(5, 7)$ has ..... value.

(a) local maximum (b) local minimum

(c) undefined (d) zero

(4) The function f has local minimum value if $f(x)$ equals .....

(a) $2x - 1$ (b) $x^2 + 1$ (c) x^3 (d) $-x^3$

(5) *(2nd Session 2024)* The local minimum value of the curve of the function $f(x) = x^2 + ax$ where a is constant at the point which lies on it and its x-coordinate equal to 3 is .....

(a) -12 (b) -9 (c) zero (d) 2

(6) If the function f has local maximum value , then $f(x)$ can be equals .....

(a) $x^2 - 3$ (b) $x^3 + 1$ (c) $x^3 + 3x$ (d) $x^4 - 3x^2$

(7) *(1st Session 2015)* If the function $f : f(x) = x + \frac{a}{x}$ has local maximum at $x = -2$, then a = ...

(a) 4 (b) 2 (c) -2 (d) -4

(8) *(2nd Session 2021)* If the function $f : f(x) = 3ax^3 - bx - 5$ has local maximum value at $x = 1$, then $\frac{b}{a} =$ .....

(a) -9 (b) 9 (c) 20 (d) -20

(9) *(2nd Session 2022)* The function $f : f(x) = x^3 + 3x^2$ has a local maximum value at the point .....

(a) $(0, 0)$ (b) $(-2, 4)$ (c) $(2, 4)$ (d) $(-2, -4)$

(10) The local minimum value of the function $f : f(x) = x \ln x$ equals .....

(a) $-e$ (b) -1 (c) $\frac{-1}{e}$ (d) 1

(11) *(1st Session 2019)* Let f be the function , defined by : $f(x) = \frac{x}{\ln x}$, then the local minimum value of f is .....

(a) e (b) $\frac{1}{e}$ (c) ln e (d) $-e$

(12) The function $f : \mathbb{R} \longrightarrow \mathbb{R}$ where $f(x) = 2x + \cos x$, then f has .....

(a) a local maximum value at $x = -2$ (b) a local minimum value at $x = -2$

(c) $\dot{f}(2) > \dot{f}(-2)$ (d) $\dot{f}(2) < \dot{f}(-2)$

(13) If f is an odd continuous function on $\mathbb{R}$ and the function has a local minimum value at $x = 2$, then the function has .....

(a) a local maximum value at $x = -2$ (b) a local minimum value at $x = -2$

(c) $\dot{f}(2) > \dot{f}(-2)$ (d) $\dot{f}(2) < \dot{f}(-2)$

(14) The curve $y = x e^{x}$ at .....

(a) $x = -1$ has a point of local minimum value.

(b) $x = -1$ has a point of local maximum value.

(c) $x = 0$ has a point of local minimum value.

(d) $x = 0$ has a point of local maximum value.

(15) If $f'(x) = (x - a)(x - b)$ where $a > b$, then all of the following statements are true except .....

(a) at $x = a$, the function has local minimum value.

(b) $\dot{f}(a) = \dot{f}(b)$

(c) at $x = b$, the function has local maximum value.

(d) $\ddot{f}(a) < \ddot{f}(b)$

(16) If $x \in]0 , \frac{\pi}{2}[$, then $f(x) = \sin\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$ has a maximum value at $x =$ .....

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{12}$ (d) $\frac{\pi}{8}$

(17) If $f'(-1) = f'(3) =$ zero and $f''(x) >$ zero for all $x \in]-2 , 2[$, then for sure .....

(a) $f(-1)$ is a local maximum value. (b) $f(-1)$ is a local minimum value.

(c) $f(3)$ is a local maximum value. (d) $f(3)$ is a local minimum value.

(18) ***(1st Session 2019)*** The function $f : f(x) = x^4 - 4x^2$ has .....

(a) one local minimum value and two local maximum values.

(b) two different local minimum values and one local maximum value.

(c) two local minimum values and no local maximum values.

(d) two equal local minimum values and one local maximum value.

(19) ***(1st Session 2022)*** If $f : f(x) = \frac{e^{2x} + 1}{e^{x}}$, then .....

(a) $(0 , 2)$ is a local minimum point of the function.

(b) $(0 , 2)$ is a local maximum point of the function.

(c) $\left(1 , e + \frac{1}{e}\right)$ is a local minimum point of the function.

(d) $\left(1 , e + \frac{1}{e}\right)$ is a local maximum point of the function.

(20) ***(Trial 2021)*** If $f(x) = \cot x - m \csc x$ has local maximum value at $x = \frac{\pi}{4}$, then the value of the constant $m =$ .....

(a) $-\sqrt{2}$ (b) -1 (c) 1 (d) $\sqrt{2}$

(21) The function f where $f(x) = (x-a)^2 + (x-b)^2$ has a local minimum value at $x =$ .....

(a) $\frac{a+b}{2}$ (b) $\sqrt{ab}$ (c) $\frac{2}{a+b}$ (d) $\frac{2ab}{a+b}$

(22) If the curve of the function f has a local minimum value at $x = a$ and $f(x) < 0$ for all values of x, then which of the following graphs has local minimum value at $x = a$?

(a) $f(x) + k$ (b) $f(x + k)$ (c) $-f(x)$ (d) $[f(x)]^2$

(23) *(1st Session 2023)* The curve of the function $xy = e^x$ has a local minimum value at $y =$ .....

(a) $-e$ (b) e (c) 1 (d) -1

(24) *(1st Session 2023)* If $f(x) = \frac{1}{x} e^{|x|}$, then the local maximum value of the function f equals .....

(a) 1 (b) e (c) $-e$ (d) -1

(25) *(Speciman 2025)* The function $f : f(x) = (x-3)e^x$ has a local minimum value at $x =$ .....

(a) zero (b) 1 (c) 2 (d) 3

4 Find the values of x which make the function has local maximum and local minimum values [where the denominator $\neq 0$] :

(1) $f(x) = x + \frac{1}{x}$

(2) $f(x) = x^2 - \frac{1}{x^2}$

(3) $f(x) = x^3 + \frac{48}{x}$

(4) $f(x) = x + \frac{4}{x-1}$

(5) $f(x) = \frac{3}{x-2}$

(6) $f(x) = \frac{x^2}{1-x}$

(7) $f(x) = \frac{1+x}{1-x}$

(8) $f(x) = \frac{x^2-3}{x-2}$

(9) $f(x) = \frac{4x-3}{x^2-1}$

(10) $f(x) = \frac{5-x}{x^2-9}$

(11) $f(x) = \frac{x^2-1}{x^2+1}$

(12) $f(x) = \frac{3x^2+5x+25}{x+2}$

5 Find the points of local maximum and local minimum of the following function :

(1) $f(x) = 3 - x^{\frac{2}{3}}$

(2) $f(x) = (x+2)^{\frac{2}{3}}$

(3) $f(x) = \sqrt[3]{x-2}$

(4) $f(x) = \sqrt[3]{(x-2)^2}$

(5) $f(x) = \sqrt[3]{27 - 8x^3}$

(6) $f(x) = \sqrt[3]{(x^2 - 9x)^2}$

(7) $f(x) = (x-1)^{\frac{4}{3}}$

(8) $f(x) = \sqrt[3]{x}\,(x-4)^2$

(9) $f(x) = \sqrt{x-3}$

(10) $f(x) = \sqrt{4 - x^2}$

(11) $f(x) = \sqrt{x^2 - 9}$

(12) $f(x) = \sqrt[3]{x^2 - 4}$

(13) 📖 $f(x) = x^{\frac{2}{3}}(2x - 5)$

(14) $f(x) = x\sqrt{x+3}$

(15) $f(x) = x^2\sqrt{x+5}$

(16) $f(x) = x^2\sqrt{3 - x^2}$

6 📖 Prove that : The function f where $f(x) = \sqrt[3]{x^2}$ has a local minimum value.

7 📖 Does the function f where $f(x) = x^3 + 3x - 4$ have local maximum and minimum values ? Explain

8 **Find the local maximum and local minimum value of each of the following functions :**

(1) $f(x) = x + 2\sin x$ where $x \in]0 , 2\pi[$

(2) $f(x) = 2x - \tan x$ where $x \in]0 , \frac{\pi}{2}[$

(3) $f(x) = \sin x + \cos x$ where $x \in]0 , 2\pi[$

(4) $f(x) = \sin x \cos x + 5$ where $x \in]0 , \pi[$

(5) *(1st Session 2008)* $y = \sin x\,(1 + \cos x)$ where $x \in]0 , \frac{\pi}{2}[$

9 **Find the local maximum and minimum values (if existed) of the function f for each of the following and show their types :**

(1) $f(x) = 4e^{-x^2}$

(2) $f(x) = e^x(3 - x)$

(3) $f(x) = x - \ln x$

(4) $f(x) = 8\ln x - x^2$

10 **Find the local maximum and local minimum value of the following functions :**

(1) $f(x) = |x + 3| - 1$

(2) $f(x) = 5 - |x - 3|$

(3) $f(x) = x|x - 4|$

(4) $f(x) = x^2|x - 3|$

11 **Find the points of local maximum values and local minimum values of the following function :**

(1) $f(x)=\begin{cases} x^2+1 & , \ x\le 0 \\ 2x+1 & , \ x>0 \end{cases}$

(2) $f(x)=\begin{cases} 3x-5 & , \ x\ge 2 \\ 3-x & , \ x<2 \end{cases}$

(3) $f(x)=\begin{cases} 3x^2+4 & , \ x\ge 1 \\ 6x+1 & , \ x<1 \end{cases}$

(4) $f(x)=\begin{cases} 3-4x-x^2 & , \ x\le 1 \\ 2x-4 & , \ x>1 \end{cases}$

(5) $f(x)=\begin{cases} 3x^2-2x-5 & , \ x\le 1 \\ x^2-5 & , \ x>1 \end{cases}$

(6) $f(x)=\begin{cases} x^3-3x^2 & , \ x\le 0 \\ x^2-2x & , \ x>0 \end{cases}$

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12 If the curve of the function $y = a x^3 + b x^2 + c x + d$ passes through the two points $(3 , 0)$, $(0 , 6)$ and it has a critical point at $(2 , 2)$

Find : a , b , c , d and determine the type of each of the critical points.

« $-1 , 5 , -8 , 6$, local maximum»

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13 If the two points $(-3 , 3)$, $(-1 , -1)$ are two critical points of the function ,

$y = a x^3 + b x^2 + c x + d$

Find : a , b , c , d and determine the type of each of the critical points.

« $1 , 6 , 9$, 3 local max. , local min.»

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14 Find the values of a , b , c , d if the curve whose equation $y = a x^3 + b x^2 + c x + d$ passes through the origin and it has a critical point at $x = 3$ and local maximum value $= f(2)$ and the tangent at $x = 1$ has equation $12x - 3y + 11 = 0$ « $\frac{2}{3} , -5 , 12 , 0$»

Higher skills

15 **Choose the correct answer from those given :**

(1) The function $f : f(x) = x^x$ has a stationary point at $x =$ .....

(a) e (b) $\frac{1}{e}$ (c) 1 (d) $\sqrt{e}$

(2) If $f : \mathbb{R}^* \longrightarrow \mathbb{R}^*$ where $f(x) = x + \frac{1}{x}$ and the function f has a local maximum value at $x = a$ and a local minimum value at $x = b$, then .....

(a) $f(a) > f(b)$ (b) $f(a) < f(b)$ (c) $a > b$ (d) $f'(a) < f'(b)$

(3) If $f : \mathbb{R} - \left\{\frac{\pi}{2} + n\,\pi\right\} \longrightarrow \mathbb{R}$ where $n \in \mathbb{Z}$, $f(x) = \tan x - x$ which of the following statements is not true ?

(a) strictly increasing.

(b) f has infinite number of critical points.

(c) f has neither local maximum nor minimum value.

(d) f has infinite number of local maximum and minimum value.

(4) If $f : \mathbb{R} \longrightarrow \mathbb{R}$ where $\dot{f}(x) = (x - 2019)\,(x - 2020)^2\,(x - 2021)^3\,(x - 2022)^2$ and $g : \mathbb{R} \longrightarrow \mathbb{R}^+$ where $f(x) = \ln g(x)$, then g has local maximum value at $x =$ .....

(a) 2019 (b) 2020 (c) 2021 (d) 2022

16 **Find the local maximum and local minimum values of each of the following :**

(1) $f(x) = |x^2 - 8x + 15|$

(2) $f(x) = \begin{cases} x^2 - 4x + 3 & , \ x \le 3 \\ 2(x - 3) & , \ 3 < x < 4 \\ |x - 6| & , \ x \ge 4 \end{cases}$

(3) $f(x) = \begin{cases} 3x - 1 & , \ x < 1 \\ 2 & , \ 1 \le x < 4 \\ 6 - x & , \ x \ge 4 \end{cases}$

17 If $f_1(x) = x^2 - 4x + 3$, $f_2(x) = 8x - 2x^2 - 9$

Prove that the two curves are tangential and find the point of tangency , then prove that it is a local maximum point to the function f_2 and in the same time its a local minimum point to the function f_1